



SAPTHAGIRI NPS UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (CSE) • TEAM INNOVEDGE
Chikkasandra, Hesaraghatta Main Road, Bengaluru, Karnataka 560057

Date: 08 October 2026

TO:

The Dean & University Administration,
Through: The Director & Faculty Mentor,
Department of Computer Science & Engineering (CSE),
Sapthagiri NPS University (SNPSU), Bengaluru – 560057

FROM:

Team InnovEdge (Dept. of Computer Science & Engineering):
• **Kiran M. Biradar** (USN: **24SUUBECS0937**) – Lead Developer & Architect
• **Karthik P.** (USN: **24SUUBECS0890**) – Co-Lead & Technical Coordinator
Under the Guidance of: **Faculty Mentor**, Dept. of CSE, SNPSU

SUBJECT: Requisition for Dedicated Server Space, Sub-Domain Provisioning, and Institutional Deployment Clearance for the 'SapthaEvent' University Management Platform.

Respected Dignitaries and Respected Faculty Authorities,

We, the student engineering leads of **Team InnovEdge** from the Department of Computer Science & Engineering, respectfully submit this official covering letter and requisition dossier for your review and administrative sanction.

Context & Reflection on Legacy 2025: As the university dignitaries will recall, an initial prototype of the campus event portal was launched during the **Legacy 2025** university festival. While the initiative demonstrated the strong campus need for a digital system, our team acknowledges with utmost professional humility and regret the operational friction encountered during that event—including server latency under concurrent loads, gate roll-call delays, and synchronization bottlenecks. We took those challenges as a profound learning mandate.

Complete Architectural Overhaul & Current State: Over the past months, under the guidance of our Faculty Mentor in the Department of CSE, Team InnovEdge has completely rebuilt, re-engineered, and hardened the entire codebase from the foundation up. We are proud to report that **SapthaEvent is now fully done, hardened, and 100% operational**, having passed **145/145 industrial automated test suites**. All past bottlenecks have been decisively solved: sub-second QR code ticket scanning with offline redundancy, dynamic department form builder, ACID-compliant database integrity, zero data loss, tamper-proof cryptographic certificates, and automatic NAAC/NIRF reporting.

Specific Requisitions Submitted for Administrative Approval:

- 1. Requisition for Dedicated Server Space:** We request allocation of dedicated server space (on-premise university server or lightweight cloud container compute) to host the production instance with zero downtime.
- 2. Official Sub-Domain Allocation:** Authorization to configure the official institutional sub-domain (events.snpsu.edu.in) with SSL intranet/internet routing for official branding.
- 3. Departmental Pilot Clearance:** Formal sanction to pilot the upgraded platform for upcoming events across the Department of CSE and all academic faculties of Sapthagiri NPS University.
- 4. Faculty Mentor & Departmental Endorsement:** Direction and backing from our Faculty Mentor and Director of CSE to onboard departmental coordinators and student volunteers.

The software entails **zero commercial software licensing cost** for Sapthagiri NPS University and is ready for immediate deployment. We humbly request the Director, Dean, and Registrar to grant favorable sanction.

Yours faithfully,

Kiran M. Biradar
Lead Developer • USN: **24SUUBECS0937**
Team InnovEdge, Dept. of CSE

Karthik P.
Co-Lead • USN: **24SUUBECS0890**
Team InnovEdge, Dept. of CSE

Faculty Mentor
Dept. of Computer Science & Engg.
Sapthagiri NPS University



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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (CSE)
Team InnovEdge • Student Technical & Software Development Council

OFFICIAL PROJECT PROPOSAL & INSTITUTIONAL REQUISITION

Comprehensive Strategic Blueprint, Architecture, and Operational Documentation for Deployment of the **SapthaEvent** Campus Event Intelligence & Orchestration Platform

DOCUMENT TITLE	Institutional Project Proposal & Server Space Requisition
PROJECT NAME	SapthaEvent — Enterprise Event Intelligence Platform
ACADEMIC SESSION	Academic Year 2026 – 2027
TARGET INSTITUTION	Sapthagiri NPS University (SNPSU), Bengaluru
DEVELOPED BY	Team InnovEdge • Dept. of Computer Science & Engineering (CSE)
PROJECT LEADS	Kiran M. Biradar (USN: 24SUUBECS0937) • Lead Developer & Architect Karthik P. (USN: 24SUUBECS0890) • Co-Lead & Technical Coordinator
FACULTY MENTOR	Faculty Coordinator • Dept. of Computer Science & Engineering
SUBMITTED TO	The Director (CSE), Dean (Engineering), and Registrar, SNPSU
PROJECT STATUS	Production Ready • 145/145 Automated Tests Passing • Fully Functional

INSTITUTIONAL ENDORSEMENT & APPROVAL MATRIX

SUBMITTED BY Team InnovEdge Leads	RECOMMENDED BY Director, School of CSE	ENDORSED BY Dean, Academic Affairs / Engg.	FINAL SANCTION Registrar, SNPSU
Kiran M. Biradar Karthik P. Team InnovEdge (CSE)	Director Dept. of Computer Science & Engg. Sapthagiri NPS University	Dean Faculty of Engineering & Tech. Sapthagiri NPS University	Registrar University Administration Sapthagiri NPS University

1. EXECUTIVE SUMMARY & PROJECT GENESIS

SapthaEvent is an enterprise-grade, institutional event management and intelligence portal engineered specifically for **Sapthagiri NPS University (SNPSU)**. Conceived and engineered indigenously by **Team InnovEdge** within the Department of Computer Science & Engineering, the portal provides a unified digital operational backbone for all technical, cultural, sporting, management, and academic events hosted across SNPSU.

Built on Python, Flask, and an ACID-compliant relational database, SapthaEvent replaces disjointed, insecure manual procedures with an automated, audit-ready platform. Following the real-world trials and operational feedback gathered during **Legacy 2025**, the software has been completely rebuilt to deliver high-speed QR check-in, real-time venue conflict detection, verifiable cryptographic certificates, competition evaluation leaderboards, and instant report generation for NAAC and NIRF accreditation.

2. THE PROBLEM STATEMENT: WHY WE BUILT SAPTHAEVENT

Sapthagiri NPS University hosts dozens of high-value technical symposia, national conferences, hackathons, sports meets, cultural fests, and workshops every academic year. However, campus event operations have historically suffered from severe administrative bottlenecks:

OPERATIONAL AREA	LEGACY / AD-HOC METHOD	SYSTEMIC RISK & INEFFICIENCY	SAPTHAEVENT SOLUTION
Student Registration	Unlinked Google Forms shared across WhatsApp groups.	Duplicate entries, lack of payment verification, spreadsheet row overwrites, and zero identity validation.	Centralized portal with SNPSU student email verification, duplicate-prevention rules, and instant database records.
Gate Access & Attendance	Physical paper clipboard registers & roll-calls at auditorium gates.	Massive entrance bottlenecks (30–45 min queues), proxy attendance, unauthorized outsider entry, safety hazards.	Sub-second digital QR code ticketing scanned via smartphone cameras with offline redundancy and duplicate detection.
Venue & Asset Booking	Informal verbal requests or written paper chits to department heads.	Severe venue clashes (auditoriums double-booked), sound system equipment contention, delayed event schedules.	Live institutional venue master catalog with real-time slot conflict alerts and automated administrative approval workflows.
Certificates & Validation	Static PNG/PDF templates exported manually via Canva/Photoshop.	Rampant certificate forgery, non-verifiable credentials, lack of permanent repository for student achievements.	Automated PDF certificates with unique cryptographic verification codes and public QR scan verification endpoints.
Competition Judging	Handwritten score-sheets calculated on paper during live fests.	Calculation discrepancies, accusations of bias, delayed prize distributions, lack of audit trail for jury scores.	Dedicated Judge Portal with multi-criteria rubric scoring (0–100), real-time leaderboards, and automatic tie-breaking.
Accreditation & Auditing	Faculty coordinators spending 2–3 weeks gathering data for NAAC/NIRF.	Missing participant demographics, unverified attendance records, loss of past event data upon student graduation.	One-click generation of comprehensive PDF/Excel reports formatted precisely to NAAC Criteria 3 & 5 and NIRF requirements.

3. THE SOLUTION: SYSTEM ARCHITECTURE & STRATEGIC VISION

SapthaEvent was engineered by Team InnovEdge to solve institutional pain points by providing an end-to-end, unified lifecycle for every event hosted at SNPSU. The architecture bridges the entire operational workflow across seven key phases:

PHASE	STAGE NAME	ACTORS INVOLVED	KEY AUTOMATION & VALUE DELIVERED
1	Event Proposal & Approval	Faculty SPOC, Director, Dean	Digital submission of event proposal, budget estimation, and venue reservation with multi-tier administrative approval.
2	Publishing & Publicity	Event Organizers, Students	Public responsive portal listing with banner, schedule, speaker bios, team size rules, and registration deadlines.
3	Registration & Ticketing	Student / External Delegate	Automated form validation, team matchmaking, coupon handling, and instant generation of cryptographic QR passes.
4	Fast Gate Ingestion	Volunteers, Security Staff	Sub-second camera scanning at entry gates. Detects duplicate scans, flags invalid tickets, and logs exact timestamp.
5	Live Competition Judging	Jury, External Evaluators	Real-time evaluation interface on tablets/phones. Multi-round rubric scoring with live leaderboard generation.
6	Credential Issuance	All Verified Attendees	System releases high-definition verifiable PDF certificates with live validation URLs only to confirmed checked-in attendees.
7	Institutional Archival	Director, IQAC, Accreditation	Automated export of attendance figures, gender ratios, participant feedback, and expense summaries for permanent archival.

KEY ARCHITECTURAL PILLARS & DESIGN PRINCIPLES

PILLAR	ARCHITECTURAL PRINCIPLE	INSTITUTIONAL VALUE FOR SAPTHAGIRI NPS UNIVERSITY
1. Zero-Trust Security	Role-based privilege isolation & strict session validation.	Prevents unauthorized student edits, protects confidential staff data, and maintains an unalterable audit log of all gate check-ins.
2. Offline Gate Continuity	Client-side caching with local synchronization queues.	Ensures check-in desks continue operating seamlessly even if campus Wi-Fi or mobile networks experience congestion during mega-fests.
3. Cryptographic Authenticity	SHA-256 digital certificate signing with public QR resolver.	Positions SNPSU as a pioneering institution with tamper-proof student credentials verifiable by global corporate recruiters.
4. Institutional Sovereignty	100% in-house software development & local database ownership.	Eliminates dependency on expensive SaaS vendors; guarantees full data privacy without third-party vendor lock-in.

4. CORE FUNCTIONAL MODULES & STRATEGIC USE CASES

The platform comprises eight purpose-built functional modules engineered by Team InnovEdge to serve the operational needs of university stakeholders:

CODE	MODULE NAME	CORE WORKFLOW & CAPABILITIES	INSTITUTIONAL BENEFIT
MOD-A	Multi-Tier RBAC & Governance	Role isolation across 6 personas: SuperAdmin (Director/Dean), Faculty SPOC, Organizer, Gate Staff, Judge, and Participant.	Eliminates unauthorized data tampering; ensures faculty retain administrative oversight of all student activities.
MOD-B	Anti-Fraud QR Ticketing	Instant dynamic QR passes sent via email/web. Camera scan check-in with duplicate detection and offline fallback.	Reduces gate entry wait times from 40 mins to 1.5 seconds per student; eliminates proxy attendance entirely.
MOD-C	Venue & Asset Booking	Institutional venue directory with slot conflict detection, capacity checks, and approval routing.	Prevents auditorium and lab scheduling clashes; optimizes utilization of university physical infrastructure.
MOD-D	Cryptographic Certificates	On-demand vector PDF certificates with unique verification hash and public scan validation URL.	Protects university brand reputation against forgery; saves over INR 50,000 annually in certificate printing costs.
MOD-E	Finance & Ledger Tracking	Tracks paid tickets, sponsor contributions, student coupons, and automated payment receipts.	Guarantees financial audit compliance; provides transparent revenue reports directly to college management.
MOD-F	Hackathon & Jury Portal	Team registration, project submissions, multi-criteria rubric scoring (0–100), and real-time leaderboards.	Ensures unbiased judging in flagship technical competitions with a fully transparent numerical audit trail.
MOD-G	Omnichannel Notifications	Transactional notifications via university SMTP / Brevo and WhatsApp API for reminders and schedule alerts.	Replaces chaotic WhatsApp group broadcasts with professional, official university communication.
MOD-H	Accreditation Reporting	One-click generation of structured reports formatted for NAAC Criteria 3.3.2 / 5.3.3 and NIRF submissions.	Saves weeks of manual administrative labor during annual inspections by maintaining real-time digital event records.

STAKEHOLDER PERSONAS & PRIVILEGE ACCESS MATRIX

USER ROLE	TARGET STAKEHOLDER	PERMITTED PORTAL PRIVILEGES	SECURITY GUARDRAIL
Super Admin	Director (CSE) / Dean / Registrar	Full platform control, departmental oversight, system audit logs, global configuration.	Master Secret Key authentication.
Department SPOC	Designated Faculty Coordinator	Approve departmental events, assign organizers, monitor venue bookings, audit budgets.	Restricted to assigned academic department.
Event Organizer	Student Council / Club Lead	Create event drafts, manage schedules, monitor registrations, allocate gate volunteers.	Requires faculty SPOC digital clearance.
Gate Volunteer	Student Event Staff	High-speed camera QR code ticket scanner, check-in dashboard, attendance logs.	Cannot view student contact PII or modify events.
Evaluator / Judge	Faculty & External Industry Jury	Access assigned team submissions, enter rubric scores, submit official rankings.	Blind scoring mode; cannot alter team rosters.
Participant	SNPSU Students & Delegates	Browse events, register, form teams, download QR passes, download verified certificates.	Self-service portal; strict data isolation.

5. TECHNICAL ARCHITECTURE, SECURITY & DATA GOVERNANCE

SapthaEvent is engineered as a robust, production-ready enterprise application compliant with industry software engineering standards, data privacy laws, and academic institutional security requirements.

ARCHITECTURAL LAYER	TECHNOLOGY SELECTION	SPECIFICATIONS & INSTITUTIONAL ADVANTAGE
Backend Core	Python 3.11 / Flask 3.x	Lightweight WSGI framework, microservices-ready, minimal server memory footprint (< 150 MB).
Database Layer	PostgreSQL (SQLAlchemy ORM)	ACID-compliant relational storage. Includes local zero-config SQLite fallback for disaster recovery and offline continuity.
Document Engine	ReportLab High-Speed Generator	Vector-grade PDF rendering engine capable of compiling thousands of certificates and reports in seconds.
Security & CSRF	Flask-WTF & Talisman Middleware	Enforces strict CSRF token validation on all mutating POST requests, secure HTTP headers (CSP, HSTS, X-Frame-Options).
Authentication	PBKDF2-SHA256 & Session Guard	Cryptographic password hashing, secure HttpOnly/SameSite cookies, brute-force rate-limiting on login endpoints.
Data Privacy	Role-Isolated PII Protection	Participant phone numbers and personal data are strictly masked from unauthorized volunteers and external attendees.
Quality Assurance	Automated Pytest Suite	145 comprehensive automated tests passing with 100% verification across security, payments, ticketing, and scheduling.

6. INFRASTRUCTURE & HOSTING COST FEASIBILITY FOR SNPSU

Because SapthaEvent is an **indigenous asset engineered by Team InnovEdge**, there are **zero commercial software licensing fees**. The financial requirements are strictly limited to the minimal server space and compute required to store and handle the data and event operations for **Sapthagiri NPS University (SNPSU)** alone (~5,000–8,000 students across all departments).

OPERATIONAL RESOURCE	COMMERCIAL SAAS (EVENTBRITE/VFAIRS)	SAPTHAEVENT FOR SNPSU	INSTITUTIONAL SAVINGS
Platform Software License	INR 2,50,000 – INR 5,00,000 / year	INR 0 (Developed In-House by InnovEdge)	100% Software License Saved
Per-Registration Commission	3% to 7% cut on all paid registrations	INR 0 (Zero platform cut for SNPSU)	100% Event Funds Retained
Server Space & Compute	Billed as part of costly enterprise tiers	Minimal Campus Server / Cloud (~INR 400/mo)	Negligible Operational Expense
Database Storage (SNPSU Data)	Hosted externally on vendor clouds	Local / Managed SQL (~300MB for SNPSU)	100% Student Privacy & Sovereignty
Certificate Generation	Third-party add-on (INR 5 – 10 / cert)	Unlimited Vector PDF Engine Included	INR 50,000+ saved on certificate printing

Summary for SNPSU Administration: Deploying SapthaEvent saves the university an estimated **INR 2,50,000 to INR 5,00,000 annually** compared to external event platforms. The total infrastructure cost to run the system across all departments of SNPSU is negligible (under INR 5,000 annually or INR 0 if hosted on internal campus servers).

7. PHASED IMPLEMENTATION & ROLLOUT ROADMAP

To ensure a seamless, non-disruptive onboarding across all academic faculties, the deployment of SapthaEvent will follow a structured 3-phase rollout plan over 6 weeks:

PHASE & TIMELINE	OBJECTIVE	ACTIVITIES & MILESTONES	DELIVERABLES / SIGNOFF
Phase 1: Weeks 1 – 2 (Sandbox & IT Integration)	Administrative Validation & Server Provisioning	• Demonstration to Director (CSE) and Dean of Engineering. • University IT mapping for sub-domain (events.snpsu.edu.in). • Provisioning of dedicated server space / container runtime. • Creation of administrative credentials for CSE faculty SPOC.	Server Space Allocation & Pilot Clearance.
Phase 2: Weeks 3 – 4 (Pilot Fest Deployment)	Departmental Pilot & Stress Testing	• Deployment for upcoming CSE departmental symposium/hackathon. • Volunteer gate check-in training (15-minute briefing). • Verification of QR ticketing flow, live judging, and certificates. • Collection of faculty and student usability feedback.	Pilot Evaluation Report & Performance Audit.
Phase 3: Weeks 5 – 6 (University-Wide Rollout)	Full Campus Institutionalization	• Onboarding of all departmental SPOCs and student clubs. • Integration into the official university academic calendar. • Centralized dashboard handoff to Dean and Campus Authorities. • Archival of all event reports for NAAC/NIRF accreditation.	Complete Institutional Operational Handoff.

8. SPECIFIC REQUISITIONS & SANCTIONS REQUESTED

For the successful deployment and institutionalization of this platform, Team InnovEdge respectfully requests formal sanction for the following items:

Requisition 1 — Allocation of Dedicated Server Space: Sanction for server compute and storage on the university data center / cloud infrastructure to host the production backend and database for SNPSU.

Requisition 2 — IT Sub-Domain & DNS Binding: Authorization for the University Systems Administrator to point events.snpsu.edu.in to the deployment server with valid SSL certificates.

Requisition 3 — Departmental Pilot Authorization: Formal permission from the Director of CSE and Dean to execute the pilot deployment for upcoming engineering symposiums and hackathons.

Requisition 4 — Notification Relay Configuration: Approval to route transactional event emails and ticket deliveries through the university's official notification relay / SMTP.

Requisition 5 — Accreditation Data Recognition: Formal recognition that event registers, verified attendance logs, and participant feedback generated by SapthaEvent constitute official institutional records for NAAC and NIRF audits.

9. INSTITUTIONAL UNDERTAKING & COMPLIANCE DECLARATION

We, the student engineering leads of **Team InnovEdge**, hereby declare and undertake the following institutional commitments regarding the maintenance and operation of the **SapthaEvent** platform:

1. **Exclusive University Ownership:** All software codebases, database schemas, documentation, and digital assets of SapthaEvent shall remain the exclusive intellectual property of Sapthagiri NPS University (SNPSU).
2. **Strict Data Ethics & Confidentiality:** No student, faculty, or institutional data will ever be monetized, shared, or transferred outside the university. All records will be securely maintained strictly within SNPSU infrastructure.
3. **Dedicated Student Maintenance & Support:** Team InnovEdge undertakes to provide continuous technical support, server monitoring, bug fixes, and volunteer training for all events without disrupting academic schedules.
4. **Zero Commercial Liability:** The platform entails no licensing or acquisition cost to the university. Any infrastructure expansions will be submitted for prior administrative review.

10. OFFICIAL SIGN-OFF, RECOMMENDATION & APPROVAL MATRIX

This project proposal and server requisition dossier has been scrutinized and is respectfully submitted for formal recommendation and executive sanction:

STAGE 1: PROJECT SUBMISSION Team InnovEdge Leads • Dept. of CSE We confirm that the software is 100% complete, fully tested (145/145 passing), and ready for university deployment. Signature 1: _____ Name: Kiran M. Biradar (USN: 24SUUBECS0937) Designation: Lead Developer & Architect Signature 2: _____ Name: Karthik P. (USN: 24SUUBECS0890) Designation: Co-Lead & Technical Coordinator Date: 08 October 2026	STAGE 2: DEPARTMENTAL SCRUTINY & RECOMMENDATION Director, School of Computer Science & Engineering I have reviewed the technical architecture, verified the resolution of legacy bottlenecks, and recommend this requisition for institutional approval. Signature: _____ Name: _____ Designation: Director, School of CSE Department Seal: [_____] Date: ____ / ____ / 2026
STAGE 3: ACADEMIC & INFRASTRUCTURE ENDORSEMENT Dean, Faculty of Engineering & Technology / Academics Endorsed for server space allocation, sub-domain binding, and campus-wide pilot deployment. Signature: _____ Name: _____ Designation: Dean, Faculty of Engineering Dean Office Seal: [_____] Date: ____ / ____ / 2026	STAGE 4: FINAL EXECUTIVE SANCTION & APPROVAL Registrar, Sapthagiri NPS University [☐] APPROVED & SANCTIONED [☐] APPROVED WITH CONDITIONS [☐] REQUIRES REVISION Remarks: _____ Signature: _____ Designation: Registrar, Sapthagiri NPS University University Seal: [_____] Date: ____ / ____ / 2026

* Official Institutional Requisition • Prepared by Team InnovEdge • Dept. of Computer Science & Engineering • Sapthagiri NPS University.